HIV 95-95-95 Dashboard: Data Cleaning Report

Prepared by Blessing Jatto | November 2026 Source: UNAIDS 2026 estimates (HIV estimates with uncertainty bounds, 1990 to 2025) | Tools: Excel, Power Query, Power BI (DAX)

1. Purpose and summary

This report records every step used to turn the UNAIDS test-and-treat workbook into one clean table for a Power BI dashboard on the 95-95-95 targets (know status, on treatment, virally suppressed). It lists what was done, why it was done, and what went wrong along the way, so the work can be repeated, checked and explained in an interview.

Starting point: a wide Excel sheet with 3 title rows, 3 header rows built from merged cells, about 100 columns and about 2,944 data rows. Values were stored as text, for example "...", ">98", "1.3 m" and "680 000".

End point: one table, HIV_Data, with 141,439 rows and 12 columns, numeric values, standard labels, flags for what was changed, and a level column that keeps Global, regions and countries apart.

Queries in the model

- Sheet1: the raw sheet (not loaded)
- Headers: a small helper that builds the column names (not loaded)
- HIV_Data: the clean table (the only one loaded)

2. The source file

File: HIV_estimates_from_1990-to-present_1.xlsx. It has four sheets:

- HIV-Test-&-Treat_ByArea: the 95-95-95 measures, Global and regions then countries, 2010 to 2025. **This is the sheet used.**
- HIV-Test-&-Treat_ByYear: appears to hold the same measures ordered by year. Not loaded, because loading both would count the same numbers twice.
- HIV2026Estimates_ByYear and HIV2026Estimates_ByArea: counts such as AIDS-related deaths. Kept for a later project on mortality.

Layout of the sheet used

- Rows 1 to 3: title, period and source text (not data).

- Rows 4 to 6: the header, split into three pieces: the indicator name, the group (All ages, Children, Adults, Women, Men) and the statistic (Estimate, Low, High).
- Row 7 onward: data. Column A is the year, column B the area code, column C the area name.
- From column D: about 97 measurement columns, repeating Estimate, Low, High for each group and indicator.
- Merged cells: a title appears once, with blanks beside it.
- Missing data shown as "...". Values above 98 shown as ">98". Counts shown as "1.3 m" (millions) or "680 000" (space as thousands separator).

3. Preparation before Power Query

1. Opened the file and clicked Enable Editing (it opened in Protected View because it was downloaded).
2. Saved a working copy as an Excel Workbook (.xlsx) with a clean file name (the & character removed). The original file was never edited. **Why:** the raw source stays available to compare against.
3. Inspected the sheets and chose one sheet, to avoid double counting.

4. Power Query steps, in order

4.1 Load the sheet

Get data > Excel workbook > tick the sheet > **Transform Data** (not Load). **Why:** to clean before anything reaches the report. Power Query added three automatic steps (Source, Navigation, Changed column type) and named the columns Column1, Column2 and so on.

4.2 Remove the title rows

Home > Remove rows > Remove top rows > 3. **Why:** the first three rows are titles, not data.

4.3 Build the Headers query

1. Right-click Sheet1 > **Reference** (creates a new query linked to Sheet1), renamed Headers.
2. Home > Keep rows > Keep top rows > 3 (only the three header rows).
3. Transform > **Transpose** (rows become columns).
4. Select Column1 and Column2 > Transform > Fill > **Down**.

5. Select Column1, Column2 and Column3 in that order > Transform > **Merge Columns**, with a custom separator, a vertical bar |, named FullHeader.

Why each step

- Reference keeps Sheet1 untouched as the raw source.
- Only three rows are flipped. The first attempt flipped the whole table and created about 2,947 columns, more than the preview could show.
- Transpose is needed because Power Query can fill down a column but has no fill right.
- Fill Down copies each merged title into its blank cells, so every column carries its full label.
- A bar is used as separator because it appears nowhere in the real text (spaces and commas do), so the name can be cut cleanly again later.

Result: 100 header names such as: Percentage of people living with HIV who know their HIV status (first 95 of 95-95-95 target) | (All ages) | Estimate. The first three rows are blank (they belong to Year, Area code and Area).

4.4 Build the HIV_Data query

1. Right-click Sheet1 > Reference, renamed HIV_Data.
2. Remove top rows > 3 (these are the header rows, already used in Headers).
3. Rename the first three columns by hand: Year, Area_Code, Area.
4. Rename the other 97 columns in one step, using a custom step (the fx button) with this formula, where the step name matches the previous step in Applied Steps:

```
Table.RenameColumns(#"Renamed columns",
List.Zip({List.Skip(Table.ColumnNames(#"Renamed columns"), 3),
List.Skip(Headers[FullHeader], 3)}))
```

How it works: it lists the current column names, drops the first three, pairs each remaining name with the matching new name from Headers (List.Zip), and applies the pairs. **Why:** typing 97 long names by hand is slow and error-prone, and this step reruns on refresh. **Risk:** pairing is by position (see section 9).

4.5 Unpivot the wide table

Select Year, Area_Code and Area > right-click > **Unpivot Other Columns**. This created two columns: Attribute (the long header name) and Value. **Why:** wide data has one column per measurement. Long data has one row per measurement, so one chart or slicer can work with any indicator, group or statistic. “Other Columns” means the identifier columns stay and everything else is melted, so a new indicator added later would also be melted.

4.6 Split Attribute into three columns

Home > Split Column > By Delimiter > Custom | > each occurrence. Renamed the results Indicator, Group, Statistic. Then Transform > Format > **Trim** on those three columns. **Why:** each column should hold one thing, so a slicer can pick “Children” or “Estimate” on its own. Trim is needed because the separator had spaces around it, and text with a leading space counts as different text.

4.7 Rename Value and set Year type

Value was renamed **Value_Raw** (the untouched text) and Year was set to Whole number. **Why:** keeping the raw text lets any number be checked against the source.

4.8 Create the numeric column, Value_Num

Add column > Custom column:

```
try
  (let
    v = Text.Trim(Text.From([Value_Raw])),
    numText = Text.Select(v, {"0".."9", "."}),
    mult = if Text.EndsWith(v, "m") then 1000000 else 1
  in
    if numText = "" then null else Number.FromText(numText, "en-US") * mult)
otherwise null
```

Rules this applies

- “...” (no data) becomes null (blank). **Missing is not zero.**
- “>98” becomes 98 (the symbol is recorded in Qualifier).
- “1.3 m” becomes 1,300,000 (m means millions).
- “680 000” becomes 680000 (the space is a thousands separator).
- Text.Select keeps only digits and the decimal point, so it works even when the space is a special non-breaking space that Replace Values cannot catch.
- “en-US” fixes how the decimal point is read, whatever the computer’s regional settings.
- try … otherwise null turns any row that cannot be calculated into a blank instead of an error. The column was then set to Decimal number.

4.9 Flag what was changed, Qualifier

```
let v = Text.Trim(Text.From([Value_Raw])) in
if v = null then "No data"
else if Text.StartsWith(v, ">") then ">"
else if Text.StartsWith(v, "<") then "<"
else if [Value_Num] = null then "No data"
else "Exact"
```

Why: “more than 98” is not the same as exactly 98, so the cleaned table keeps the honest label.

4.10 Separate percentages from counts, Unit

```
if Text.StartsWith([Indicator], "Percentage") then "Percent" else "Count"
```

Why: a chart must never mix 88 (a percentage) with 29,800,000 (people) on one axis.

4.11 Standardize the Group labels

Transform > Replace Values: replaced “(” with nothing, then “)” with nothing, so “(All ages)” and “All ages” became the same label. **Why:** to a computer these are different texts, so a slicer would list the same group twice.

4.12 Remove rows with no value

Filtered Value_Num to remove nulls (this also removed the “...” rows). **Why:** null already means “not reported”, and removing these rows makes the table lighter. **Note for the record:** write down the row count before and after this step (before: _____, after: 141,439).

4.13 Short names, Indicator_Short

A custom column that maps the very long indicator names to short ones:

```
let i = [Indicator] in
if Text.Contains(i, "first 95") then "1st 95: Know status"
else if Text.Contains(i, "second 95") then "2nd 95: On treatment"
else if Text.Contains(i, "third 95") then "3rd 95: Viral suppression"
else if Text.StartsWith(i, "Percentage") and Text.Contains(i, "pregnant")
then "% pregnant women treated"
else if Text.StartsWith(i, "Percentage") and Text.Contains(i,
"antiretroviral") then "% on treatment (all with HIV)"
else if Text.StartsWith(i, "Percentage") and Text.Contains(i, "suppressed")
then "% suppressed (all with HIV)"
else if Text.StartsWith(i, "Number") and Text.Contains(i, "pregnant") then
"Pregnant women (count)"
else if Text.StartsWith(i, "Number") then "People with HIV (count)"
else i
```

Why: the long names do not fit in chart titles or slicers. **Important:** the three 95 measures are percentages of the previous step. The “(all with HIV)” measures use everyone living with HIV as the base, so they are different measures and are never mixed with the three steps.

4.14 Separate Global, regions and countries, Area_Level

```
if [Area_Code] = "03M49WLD" then "Global"
else if Text.StartsWith([Area_Code], "UNA") then "Region"
else "Country"
```

Why: adding Global to its regions would double count. Each visual uses one level only.

4.15 Load settings

Unticked **Enable load** for Sheet1 and Headers (Power BI warns that visuals using them would break, which is safe because none do), then Home > Close and Apply. Only HIV_Data loads. **Why:** the raw sheet and the header helper are scaffolding, not report data.

5. Validation checks

- **Row count:** HIV_Data shows 141,439 rows. The maximum possible is about 285,500 (2,944 rows × 97 columns), and a lower number is expected because blank cells were removed.
- **Spot checks against Excel (Global, 2025, All ages, Estimate):** 1st 95 = 88 and 2nd 95 = 89 agreed between the Excel sheet and the dashboard cards. The 3rd 95 card shows 95. Result for the 3rd 95 to confirm in Excel: _____.

- **Duplicates:** select Year, Area_Code, Indicator, Group, Statistic, then Keep duplicates. The expected result is zero rows. **The Keep duplicates step must be deleted afterwards**, because left in place it removes all other rows. Result: _____.
- **Range checks:** percentages between 0 and 100; years 2010 to 2025 only; Statistic holds only Estimate, Low and High; Low ⩽ Estimate ⩽ High. Results: _____.
- **Area_Level:** Global, Region and Country each contain the expected areas, and no region is coded as a country. Result: _____.
- **Dashboard against source:** every card and chart point is compared with the same cell in Excel. A mismatch between a card and a chart led to a filter error being found (section 7, item 9).

6. Measures used by the dashboard

Written in DAX, on one line each. All of them use only Estimate rows for the headline numbers.

```
Estimate % = CALCULATE(AVERAGE(HIV_Data[Value_Num]), HIV_Data[Statistic] =
"Estimate", HIV_Data[Unit] = "Percent")

Latest Estimate % = CALCULATE([Estimate %], LASTNONBLANK(HIV_Data[Year],
[Estimate %]))

Gap to 95 = 95 - [Latest Estimate %]

Low % = CALCULATE(AVERAGE(HIV_Data[Value_Num]), HIV_Data[Statistic] = "Low",
HIV_Data[Unit] = "Percent")

High % = CALCULATE(AVERAGE(HIV_Data[Value_Num]), HIV_Data[Statistic] =
"High", HIV_Data[Unit] = "Percent")

YoY change = VAR CurrentYear = MAX(HIV_Data[Year]) VAR Prior =
CALCULATE([Estimate %], HIV_Data[Year] = CurrentYear - 1) RETURN
IF(ISBLANK(Prior), BLANK(), [Estimate %] - Prior)
```

(If the Latest Estimate % formula used in the file differs from this one, replace it here with the exact version that is in the file.)

Rules for every visual: filter to one Area_Level; filter Group (usually All ages); filter Indicator_Short to the step shown. Without these filters a visual averages different things together and the number means nothing.

7. Problems met and how they were fixed

1. **Transposing the whole table** produced about 2,947 columns and a truncation warning. Cause: transpose turns every row into a column. Fix: undo the step, and flip only the three header rows in a separate query.
2. **Merge separator typed as a description.** The first attempt produced the words "(space, bar, space)" in every header, and the second "a space, then the bar…". Cause: the separator box inserts exactly what is typed. Fix: edit the Merged Columns step (gear icon) and enter only the bar character.
3. **Error "Token Eof expected"** in the rename formula. Cause: a stray closing quotation mark left after pasting. Fix: delete the extra character.
4. **Error "Token Literal expected".** Cause: the formula started with an equals sign, which this formula bar does not use. Fix: paste the formula without the leading =.
5. **[Error] in Value_Num and Qualifier** for the "..." rows. Cause: text with no digits could not be converted, and one reference used the wrong capital letter (column names are case-sensitive in Power Query). Fix: the try … otherwise null version, and the exact name Value_Raw.
6. **A leftover Kept duplicates step** would have thrown away every non-duplicate row. Fix: delete check steps after using them.
7. **"Possible data loss" warning** when unticking Enable load. Cause: Power BI warns every time. Fix: continue, because no visual used those queries.
8. **Measure named "VAR LatestYear"** and cards showing 202.50K%. Cause: a multi-line paste lost the first line, so Power BI took the VAR line as the measure's name and the measure returned the year (2025). Fix: delete it and write the measure on a single line, with the name before the first equals sign.
9. **Card said 88 but the chart said 78.** Cause: the page filters (Area_Level = Global, Group = All ages) had been removed, so the chart averaged all areas and groups. Fix: restore the page filters, then compare cards, chart and Excel.
10. **YoY formula error.** Cause: a MAX calculation was placed directly inside a filter, and a bracket was doubled when typed. Fix: store the year in a VAR first, and paste instead of typing.
11. **Map without colour saturation.** Cause: this version of the filled map has no such field. Fix: Format > Fill colours > Default colour > fx > gradient on Latest Estimate %, minimum 50 and maximum 95, with one step selected.

8. Decisions and assumptions

- "..." is treated as missing and left blank, never zero.
- ">98" is stored as 98 with a ">" flag, so it is a lower bound, not an exact value.

- “m” means millions. These counts are rounded (for example to 100,000), so they are approximate.
- Estimate, Low and High stay as separate rows, so one measure can show an estimate and its range.
- Percentages and counts are separated with Unit.
- Global, regions and countries are never added together.
- Rows without a value were removed. The 95-95-95 indicators have no values for 2010 to 2014, so those series start in 2015.
- Only one source sheet was used, to avoid double counting.

9. Known limitations

- **Position-based renaming:** if UNAIDS adds or reorders columns, names could land on the wrong data. Keep the raw file, check a few columns after any refresh.
- **Capped values:** values shown as “>98” hide the true number, and the flat 98 line in the 3rd 95 range chart comes from this cap.
- **Wide uncertainty:** some estimates (for example the 3rd 95 globally) have a wide Low to High range, so a value at target can still be uncertain.
- **Averaging:** average across countries is unweighted. Use the published regional rows for regional comparisons.
- **Dates:** the data ends in 2025 and some indicators start later than 2010.

10. How to repeat or refresh the work

1. Replace the source file and choose Home > Refresh.
2. Open Power Query and check the Applied Steps for errors, and check the count of columns (the rename step assumes 100).
3. Re-run the validation checks in section 5.
4. Compare Global 2025 values with the source sheet.

11. Skills shown (interview talking points)

- **Multi-row headers:** how they were rebuilt with Reference, Keep top rows, Transpose, Fill Down and Merge.
- **Tidy data:** why wide became long, and why one column holds one thing.
- **Defensive formulas:** try … otherwise, and treating missing as blank rather than zero.
- **Honest data handling:** flags for “>98”, notes on rounded counts, and showing uncertainty.

- **Validation:** comparing the dashboard to the source, and finding the filter error through a mismatch.
- **Debugging:** reading the error message, finding the cause, and fixing one thing at a time.

12. Short version for the README

Data cleaning (Power Query): rebuilt three-row merged headers into single column names, unpivoted 97 measurement columns into long format, split names into indicator, group and statistic, converted text values such as “1.3 m” and “>98” to numbers with a qualifier flag, treated “…” as missing (not zero), separated percentages from counts and Global from regions and countries, and validated the result against the source file (141,439 rows loaded).